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# Quantum communication and computing: Elevating the banking sector

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Quantum computing is delivering its first real-world benefits in financial services, realizing early business value today.

**Quantum computing and communications** have recently seen breakthroughs in fault-tolerant computing and long-distance secure communication. Ambitious road maps from major players suggest that "Q-Day"—the future date when quantum technologies demonstrate a definitive real-world advantage—may arrive earlier than expected. However, this acceleration also signals that the day quantum computers may break current security algorithms is approaching.

Quantum computing represents a new paradigm in calculation, utilizing the principles of quantum physics to solve complex problems rapidly. It excels at tasks currently too intricate or time-consuming for even the most powerful classical computers. The technology is still maturing, but leading financial institutions are already preparing for the associated

cyber risks by exploring its capabilities while demonstrating how they can deliver significant business advantages with the technology.

Although challenges remain in scaling up quantum computing, financial institutions are shifting from creating proofs of concept to codeveloping use cases with specialized quantum players. Hybrid computing—that is, running specific parts of a workflow on a quantum computer while processing the rest classically—allows institutions to harness near-term business value while using currently available hardware. This article explores how banks are protecting themselves against emerging risks and leveraging quantum capabilities to generate new value.

# Cyber risks can be managed through quantum communication

While quantum computing can enhance financial performance, it poses a significant security threat. A sufficiently powerful quantum computer could break the cryptographic algorithms that rely on mathematical problems too difficult for classical computers to solve. Fortunately, solutions are being developed through changes in underlying security algorithms, such as postquantum cryptography (PQC), and the advent of new technologies, such as quantum key distribution (QKD).

# Postquantum cryptography

Current public-key cryptography relies on mathematical problems that would be easily solvable by a quantum computer. PQC utilizes a new class of mathematical problems believed to be sufficiently complex to resist quantum computational advantages. Because PQC does not require specific hardware changes, banks can adopt it in the near term.

Recognizing this imperative, most leading financial institutions are actively deploying new PQC solutions. They are also prioritizing “crypto agility”—the ability to rapidly update hardware and software cryptographic systems—to adapt quickly to evolving threats and standards.

# Quantum key distribution

QKD is a mathematically provable, quantum-secure solution based on quantum hardware. Leveraging the laws of quantum physics, QKD exchanges quantum keys during the communication process; any attempt at eavesdropping alters the quantum state and is immediately detected. By nature, this concept is resistant to attacks by quantum computers. However, compared with PQC, QKD’s total cost of ownership is higher and deployment times are longer because it requires specific hardware.

While regulators are currently focused on near-term PQC, QKD is essential for long-term, comprehensive quantum security. Financial services have stringent cryptographic requirements, and to prepare for Q-Day, they must aim for high crypto agility. IT modernization efforts are a critical enabler of this agility.

As part of OpenQKD, an EU-funded initiative to build a quantum communications infrastructure across Europe, financial institutions are piloting QKD to secure critical data transfers. For example, Danske Bank in Denmark successfully completed a live QKD-protected transfer between simulated data centers. This represented the first quantum-safe data exchange in the Nordics outside a lab environment.[1] These pilots—advanced by institutions including ID Quantique and DTU (the

Technical University of Denmark)—demonstrate QKD's potential to future-proof financial communications.

Similarly, HSBC has partnered with Quantinuum to explore how quantum computing can enhance the security of digital assets and distributed ledger systems. HSBC is testing quantum-generated cryptographic keys to secure tokenized gold transactions on its Orion blockchain platform.[2] These keys function similar to a lock that randomly and continuously changes its combination, making them exponentially harder for attackers to breach than static encryption methods, even with future quantum computers. Early results indicate that quantum-safe encryption can be seamlessly integrated into existing blockchain systems without disruption, offering a practical path to protecting digital assets against future threats.

Data security in the quantum era necessitates rethinking current cybersecurity approaches. Because sensitive data sets can be "harvested" today and decrypted later, banks must identify and protect critical data now. Quantum communication addresses this threat by physically securing data transmission, providing a long-term defense against future decryption capabilities.

# Quantum computers can enhance financial performance

Major business units within financial institutions manage compute-intensive tasks that are well-suited to quantum and hybrid computing. By adopting hybrid approaches, institutions can solve complex problems today without waiting for fully scaled quantum hardware to mature. The potential economic value of quantum computing in the

finance industry is estimated to reach between $400 billion and $600 billion by 2035.[3]

As hybrid computing enables short-term value generation, financial players are increasing their investments in the space, exploring use cases in various business units. While scientific publications often categorize use cases by the mathematical problem they address, such as optimization, simulation, machine learning, or prime factorization, this article highlights applications from a business perspective. The following use cases address one or more of three levers: cost reduction, revenue enhancement, and new business opportunities. The last remains underestimated, but it possesses tremendous long-term potential to reshape the industry.

# Derivative pricing

Derivative pricing involves determining the value of financial instruments based on an underlying asset. Quantum computing enables consideration of a broader set of input factors than classical approaches, improving the accuracy and speed of pricing algorithms.

# Collateral optimization

Quantum computing offers a powerful approach to optimizing collateral allocation—a complex matching problem between assets and credit lines, particularly for corporate clients. Quantum algorithms can solve this efficiently while respecting eligibility, concentration, and liquidity constraints.

# Credit risk evaluation

Credit risk evaluation requires analyzing various parameters to assess the risk of lending to a customer and adjusting the parameters of the loan accordingly. Quantum computing enables institutions to consider significantly more scenarios than a classical computer can calculate within a practical time frame. It is also more efficient for calculating essential metrics, such as economic capital requirements for a bank's full portfolio—which are vital for financial stability—and offers superior advice on individual investment decisions.

For instance, the Fidelity Center for Applied Technology collaborated with IonQ to develop and train quantum models that generate realistic synthetic financial data.[4] These models accurately reflect complex market behaviors and intervariable relationships, producing financial data that are more realistic and accurate than the data produced by traditional methods. This advancement enables improved testing and validation of financial models, helping institutions refine credit risk assessments.

# Fraud detection

Quantum computing can significantly enhance fraud detection through quantum machine learning, enabling the rapid, precise analysis of large, complex transaction data sets. By improving the speed and accuracy of identifying subtle patterns and anomalies, quantum computing detects fraud within the narrow time frames required for transactions. This enhances security for institutions and customers while reducing false positives.

Intesa Sanpaolo, a major Italian banking group, is collaborating with IBM to explore quantum machine learning for this purpose.[5] The bank uses a quantum algorithm to classify and identify data patterns that are too complex for traditional methods. In initial tests, the quantum model identified fraudulent transactions with greater accuracy and efficiency, reducing the number of legitimate transactions flagged as false positives.

## Churn prediction

Quantum computing offers powerful capabilities for predicting churn in retail banking. By leveraging quantum algorithms, institutions can more accurately identify customers at risk of leaving and gain deeper insights into the underlying reasons. This enhanced predictive power enables banks to implement more-effective retention strategies, improving loyalty and safeguarding revenue.

One example is Itaú Unibanco's collaboration with QC Ware, which demonstrated the use of quantum-inspired algorithms to improve financial forecasting, specifically to reduce customer churn.[6] Applied to a data set of approximately 180,000 anonymized customer records, the model improved overall precision by 8 percent and increased the detection of customer withdrawals by 2 percent.

## Quantum money

"Quantum money" refers to a form of digital currency secured by the laws of quantum physics. This would allow for unforgeable, verifiable, private transactions—covering currency and other assets—thereby

enhancing financial security and trust. Proposals and initial proof points for quantum-digital payment methods have already been published.[7]

# Actions for financial services to stay prepared and capture early value

Banks can take six specific actions to position themselves for the transformative quantum era:

1. **Prepare for quantum security.** Assessing a bank's security posture and identifying sensitive data sets are key prerequisites for quantum preparedness. Early engagement with relevant companies in the PQC and QKD space will help identify appropriate technological solutions for Q-Day.

2. **Develop a clear strategy.** Financial services firms should articulate a clear quantum strategy, ideally covering a three-year horizon. This begins with defining goals and prioritizing the actions required to deliver them. Crucially, institutions must determine organizational responsibilities and how to integrate quantum capabilities with existing computing infrastructure.

3. **Assess value throughout business units.** Evaluating the value of compute-intensive tasks across a bank helps set the correct focus for capturing near-term value. Starting with "low-hanging fruit" and continuously reinvesting gains allows for a sustainable, long-term approach.

4. **Build quantum capabilities.** While quantum computing will become more accessible over time, dedicated internal capabilities are a prerequisite for active entry into the space. This can be achieved through multiple avenues:

- partnering with quantum players that bring external capabilities, consulting, and education
- upskilling the internal workforce—particularly tech talent—to develop quantum algorithms and run them on available hardware
- hiring specialized talent to work closely with business departments on use case implementation

5. **Implement use cases.** Beyond hardware and software development, a crucial step toward adoption is bridging the gap between quantum technology capabilities and business expertise. Close collaboration between tech departments, quantum players, and business units helps institutions tackle the problems that matter most. While pilots provide early indications of value, long-term buy-in and the codevelopment of use cases offer the best chance for success.

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Banks can take action today to prepare for and actively shape the quantum era. Acting now allows financial services players to capture early value, secure intellectual property, and derisk their cryptographies. By developing a clear strategy, assessing value across business units, building capabilities, and implementing use cases, financial services can position themselves to thrive in—and shape—this transformative future.

***Editorial note***

*This article was updated in February 2026; it was originally published in November 2025. Some data in this article have been updated to reflect evolving analysis.*

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